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PALEOMAGNETISM OF THE CAMBRO-ORDOVICIAN ARBUCKLE GROUP AND ASSOCIATED DEPOSITS IN THE SOUTHERN OKLAHOMA AULACOGEN: EVIDENCE FOR BLOCK ROTATIONS

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Abstract. The magnetization in hydrocarbon-impregnated Arbuckle Group limestones from the southern margin of the Southern Oklahoma Aulacogen (Limestone Hills) is secondary and resides in magnetite. The magnetization (Declination/Inclination= 128°/2°, α_{95}=7.6°, k=79) fails a fold test and is post Pennsylvanian. The corresponding pole position (146°E, 30°N), however, does not coincide with the post-Pennsylvanian Apparent Polar Wander Path (APWP). Some Permian speleothems that fill caves in these rocks contain magnetizations that are also anomalous. Shallow remanent inclinations from the folded limestones and speleothems suggest that the most likely time of remanence acquisition was the Permian. The directions are at least 30° in declination from the expected directions based on the inferred ages of the magnetizations. This discrepancy can be explained by a horizontal counterclockwise rotation, consistent with left-lateral wrench faulting, during deformation of the aulacogen.

Introduction

The Cambro-Ordovician Arbuckle Group is a thick sequence of shallow-water limestones which is exposed in southern Oklahoma. Previous paleomagnetic results from the unit have documented the presence of late Paleozoic magnetizations that reside in hematite [e.g. Elmore et al., 1985; Cochran and Elmore, 1987]. The current study was designed to sample hydrocarbon impregnated limestones in the unit to test for a relationship between hydrocarbons and authigenic magnetite. The results of the study have implications for this problem, but are also of interest because they provide evidence for block rotations associated with left-lateral wrench faulting in the Southern Oklahoma Aulacogen.

The study area is located in the Limestone Hills, on the southern margin of the Southern Oklahoma Aulacogen (Figure 1). The Limestone Hills are bounded on the south by the Meer's fault and Wichita Mountains, and on the north by the Mountain View fault and Anadarko Basin. Two structural terranes separated by the Blue Creek Canyon fault are exposed in the Limestone Hills. North and east of the fault lies the Blue Creek Horst, which consists of homoclinally tilted rocks [Donovan, 1986]. To the south and west of the Blue Creek Canyon fault lies the Lawtonka graben, which consists of complexly faulted and folded rocks.

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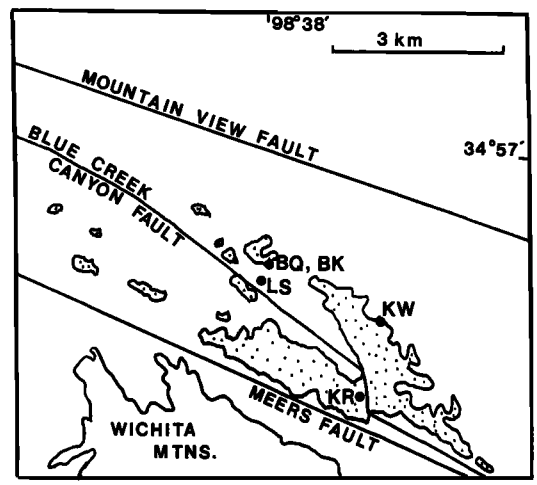


Fig. 1. Study area wtih major tectonic elements and sampling locations (KR, KW, BQ, BK, LS).

Pennsylvanian deformation in the southern Oklahoma aulacogen is characterized by several structural styles [Donovan, 1986]. For example, compression is indicated by reverse faults and tight folding [Brewer et al., 1983]. Wickham [1978] cites the trends of the folds and faults, and other structural evidence from the region, in support of left-lateral wrench movement along faults parallel to the trend of the aulacogen. Donovan [1986] argues that a left-lateral transpression model should be applied to the area.

Previous paleomagnetic work in the Limestone Hills focused on speleothems found in Ordovician age host rocks [Elmore et al. 1987]. The speleothems are Permian in age based on interbedded fossils and contain a Permian direction of magnetization that is interpreted to have been chemically acquired soon after deposition. The host rocks of the speleothems are stratigraphically equivalent to Ordovician rocks in the Arbuckle Mountains 50 km to the east, which possess late Pennsylvanian-Permian magnetizations residing in hematite [Steiner, 1973; Elmore et al., 1985; Cochran and Elmore, 1987].

Samples were collected with a portable coring apparatus and oriented in place with a clinometer

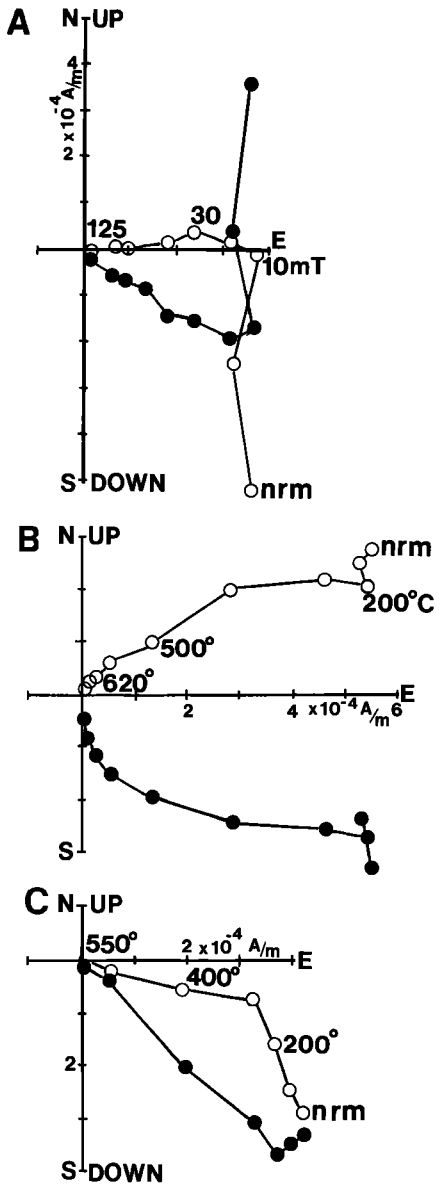


Fig. 2. Orthogonal projections illustrating demagnetization of representative specimens. A) AF demagnetization of KR4-4 showing removal of a northerly and steep down modern component at low field strengths followed by a southeasterly and shallow component at higher field strengths. B) Thermal demagnetization of BQ4-28 illustrating curved path. C) Thermal demagnetization of BK1-2 showing removal of a southeasterly and shallow component. All plots in before-tilt coordinates. Closed symbols, projection on horizontal plane; open symbols, projection on vertical plane.

and compass. The samples were collected from five locations (Figure 1) and several different rock types. Arbuckle Group limestone beds were sampled at the Paradox Anticline (location KR) in the Lawtonka graben to perform a fold test. Calcite speleothem deposits, some of which are horizontal and are paleodip indicators, were sampled at three localities (KW, BQ, LS) in the Blue Creek horst. At localities KW and BK shallowly dipping carbonates of the Kindblade Formation (Upper Arbuckle Group) were also sampled. Results of organic geochemical analyses indicate that all rocks sampled contain hydrocarbons.

Remanence measurements were made on an ScT cryogenic magnetometer. Specimens were demagnetized in a stepwise fashion using either alternating field (AF) techniques in a shielded two axis tumbling unit or thermal techniques in a Schonstedt TSD-1 unit. Directional and intensity results were plotted as orthogonal projections. Best fit demagnetization lines and planes were calculated for linear and planar demagnetization trajectories using principal component analysis [Kirschvink, 1980]. Representative specimens were selected for rock magnetic experiments to aid in identification of the magnetic mineralogy.

RESULTS

AF and thermal demagnetization of specimens from the Paradox Anticline (location KR) typically removes a southeasterly and shallow direction (Figure 2a). In some specimens a northerly to northeasterly and steep down component, interpreted to be of recent (viscous) origin, was removed at low field strengths or temperatures. The site-mean directions for the southeasterly component group well before tilt correction and diverge after tilt correction (Figure 3a; Table 1). Although the grouping of the site-mean directions slightly improves with partial unfolding, the improvement is not significant at the 95% confidence level [McElhinny, 1964]. The 0% and 100% tilt corrected means are significantly different at the 99% confidence level. These results indicate that the magnetization is post folding.

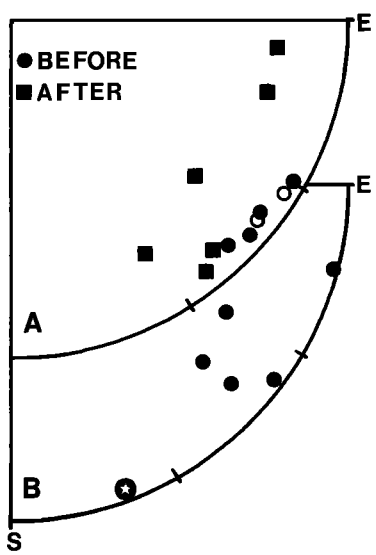


Fig. 3. Equal-area projections. A) Site means from Paradox anticline (KR) for before and after tilt correction. B) Mean directions for all locations and the expected Permian direction (star) from Elmore et al. [1987]. Open symbols, upper hemisphere; solid symbols, lower hemisphere.

Samples collected from location KW include a variety of lithologies: oosparite, pelsparite, dolomite, and speleothem. The magnetization in specimens from all sites at KW decays in a

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Table 1. Paleomagnetic data

Site	n/n_0	Decl/Incl (in situ)	k	α_{95}	Decl/Incl (corrected)
KR1 (1)	5/6	136/ 8	1008	2.4	105/22
KR2 (1)	6/6	128/ 8	142	5.6	95/22
KR3 (1)	4/5	132/ 5	69	11.2	130/30
KR4 (1)	4/4	122/-5	116	8.5	139/10
KR5 (1)	6/6	120/ 4	73	7.9	152/21
KR6 (1)	3/6	129/-8	41	19.6	142/ 6
*Mean for KR in situ	6/6	128/ 2	79	7.6	---
**Mean for KR corrected	6/6	---	13	19.6	128/20
BQ4 (s)	24/28	104/ 0***	--	--	---
LS (s)	7/7	132/13	29	11	---
Mean for KW(1) sites	5/5	120/27	37	12.6	117/23
Mean of BK(1) sites	5/5	131/23	38	12	121/21

Note: Decl = declination, Incl = inclination. k is Fisher's precision parameter; α_{95} is radius of cone of 95% confidence. n/n_0 is ratio of specimens/sites used in statistical calculations to specimens/sites analyzed. (s)= speleothem site, (l)= limestone site.
*Pole position for the in situ direction is N146.8°E, 29.5°N.
**Corrected data includes a 25° plunge correction applied in a stepwise fashion during unfolding.
***Q value for BQ4 remagnetization circle data is 0.0076.

southeasterly and shallow to moderate direction during AF and thermal demagnetization (Figure 3b; Table 1). Directions for the Kindblade limestone at location BK, the host rock for the BQ speleothems, are southeasterly and shallow to intermediate (Figures 2c,3b; Table 1). The speleothem at location LS also contains a southeasterly and shallow-to-intermediate direction of magnetization (Figure 3b; Table 1).

The previously published paleomagnetic study of hydrocarbon-saturated calcite speleothem deposits at location BQ documented that they possess a southeasterly and shallow (decl.=159°, incl.=+3°,k=29, α_{95}=5°) direction of magnetization that indicates remanence acquisition in the Permian [Elmore et al., 1987]. A more recent study of one isolated but lithologically similar speleothem deposit (site BQ4) suggests a more complex magnetization history.

Demagnetization trajectories for most specimens from site BQ4 are curved, commonly changing from a southeasterly direction to a south-southeasterly direction during progessive treatments (Figure 2b). This suggests simultaneous removal of two (or more) components. The magnetization in a few specimens, however, exhibited linear decay in a east-southeasterly and shallow direction and the magnetization in one specimen decayed in a south-southeasterly and shallow direction.

Best-fit planes were calculated for the demagnetization data from specimens from BQ4 with curved paths and were analyzed using remagnetization circles [Halls, 1976]. The great circles converge at an easterly and shallow direction (Figure 3b; Table 1). The intersection corresponds to a direction common to all samples, the component with the least dispersion. The trends of the demagnetization trajectories suggest that the other unresolved component is south-southeasterly and shallow, and probably corresponds to the Permian direction found in the other speleothems at location BQ.

Maximum unblocking temperatures for specimens from all locations generally ranged between 500 and 575°C, although a few went as high as 620°C. IRM acquisition experiments indicate the presence of a low coercivity phase in all specimens examined, with some samples containing a higher coercivity phase such as hematite. IRM decay indicates that at least 95% of the IRM is removed by 580°C. The AF and thermal demagnetization results, and the IRM experiments suggest that the dominant magnetic mineral that carries the remanence is magnetite. Magnetic spheres, similar to reported authigenic magnetite spheres [e.g. Elmore et al., 1987] are also found in the limestones.

DISCUSSION

The results of the fold test from location KR indicate that the magnetization is post folding, but the pole position that corresponds to the magnetization does not fall on the post-Pennsylvanian APWP. The shallow inclination for the post-folding direction suggests that the most likely time of remanence acquisition is in the Permian. The mean declination for the specimens from the anticline is approximately 30° in a counterclockwise sense from an expected late Pennsylvanian-Permian direction (Figure 3b). The same arguement applies to the speleothem deposits; they are Permian in age, but the directions are at least 30° in declination from the expected direction (Figure 3b).

The discrepancy between the measured directions and the expected directions based on the inferred age of remanence acquisition can be explained by a rigid body tectonic rotation around a vertical axis. As previously mentioned, the study area is thought by some to have undergone left-lateral wrench faulting and/or transpression in the late Paleozoic. This could have rotated the observed magnetic directions into their more easterly positions (Figure 3b).

Limestones from localities BK and KW have declinations which are similar to those from the Paradox Anticline, although the inclinations are steeper and the results are not constrained by fold tests (Figure 3b, Table 1). The declinations from the speleothems at locations LS and KW also suggest a rotation of approximately 30°.

The results from the speleothem at site BQ4 indicate simultaneous removal of at least two components of magnetization. The easterly and shallow intersection of remagnetization circles apparently represents one component. The easterly declination for these Permian speleothems suggests rotation of this component. The other unresolved component is interpreted to be south-easterly and shallow based on demagnetization trajectories. The speleothem at site BQ4 acquired part of its total magnetization (the now easterly and shallow component) before rotation, and part (represented by the unresolved but probably south-southeasterly and shallow component) presumably after rotation. Results from site BQ4 suggest a larger rotation (50°) than other sites

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(Table 1; Figure 3b). The differences could be explained by acquisition of magnetization during different stages of rotation.

Although simultaneous removal of components is clearly the case in samples from site BQ4, we do not think this is a problem at the other sites. Magnetizations in the other specimens decay in a linear fashion. The consistency of the results from the limestones and the Permian speleothems (BQ4) also supports the interpretation that the directions are rotated Permian components and are not due to vector addition.

Although the paleomagnetic evidence indicates that crustal blocks have undergone counterclockwise rotations, there is still uncertainity about the nature of the tectonic processes involved. Two models for rotational kinematics have been proposed to explain large horizontal rotations of crustal blocks within zones of strike-slip faulting. In one model local faults, with an opposite sense of movement to that on the major wrench system, separate elongate crustal slivers [e.g. Nur et al., 1986]. A second model calls for blocks to rotate like balls in a bearing [e.g. Brown and Golombek, 1985]. The oblique orientation of some north-northwest–south-southeast faults between the major, northwest–southeast trending Meers and Mountain View faults is consistent with the crustal sliver model. There are no reports, however, of any corroborating evidence for right-lateral movement along these faults as predicted by the sliver model.

The magnetization in the speleothems and limestones is secondary and predominantly resides in magnetite. A thermoviscous origin for the magnetization can probably be eliminated because of low burial temperatures for the rocks in the area [Cardott and Lambert, 1985]. The magnetization is, therefore, interpreted to be chemical in origin. Elmore et al. [1987] report that the speleothems with primary, hydrocarbon-filled fluid inclusions contain a magnetization that resides in authigenic magnetite whereas speleothems without hydrocarbons contain a weak and unstable magnetization. This evidence suggests that the chemical conditions created by the hydrocarbons caused percipitation of authigenic magnetite and acquisition of an associated remanence. The magnetization in the limestones may also have been acquired when hydrocarbons seeped into the rocks and created the chemical conditions necessary for authigenic magnetite to precipitate.

In summary, the paleomagnetic results suggest that block rotations of 30-50° occurred in the Limestone Hills. The rotations are consistent with left-lateral wrench faulting or transpression in the area during deformation of the Southern Oklahoma Aulacogen in the Pennsylvanian-early Permian. Because the rotation is younger than the folding, the results are also consistent with compression followed by left-lateral wrench faulting as suggested by some authors [e.g. Brewer et al., 1983].

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